

Subject Code	AF5637
Subject Title	Microeconomics
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Basic knowledge of single-variable calculus is presumed.
Objectives	The course contributes to Program Learning Outcome 1 by providing students with a rigorous introduction to microeconomic theory and applications that sit at the core of modern economic science. The course is designed with a dual focus on building economic intuition as well as equipping students with the fundamental mathematical tools commonly used in economic analysis.
Intended Learning Outcomes	Upon completion of this course, students will: 1) gain a solid understanding of modern microeconomic theory 2) develop strong economic intuition, as well as analytical and critical thinking skills 3) be equipped with fundamental mathematical tools in economics needed for further studies 4) be able to design, set up, and solve quantitative economic models 5) be able to apply microeconomic theory in other fields in economics, and to explain and critically analyze economic and social phenomena in the real world
Subject Synopsis/ Indicative Syllabus	1) Mathematics for Microeconomics a. Basic multivariable calculus b. Static optimization c. Constrained optimization and inequality constraints d. Implicit functions and the envelope theorem 2) Consumer Theory a. Preferences and utility b. Utility maximization c. Income and substitution effects d. Consumer demand 3) Producer Theory a. Production functions b. Cost functions c. Profit maximization d. Output supply and input demand 4) Partial equilibrium 5) Imperfect Competition

	a. Monopoly b. Oligopoly 6) Game Theory a. Nash equilibrium b. Mixed strategies c. Sequential and repeated games 7) Uncertainty and Information a. Expected utility theory b. Choice under uncertainty c. Asymmetric information d. Principal-agent model e. Moral hazard f. Introduction to mechanism design																																																																						
Teaching/Learning Methodology	There will be lectures each week focusing on conceptual matters, which will be supplemented by in-class tutorial where worked examples will be delivered to consolidate the theoretical concepts. Students are expected to self-study outside of class time by reading the textbook and solving practice problems.																																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>(1)</th><th>(2)</th><th>(3)</th><th>(4)</th><th>(5)</th><th></th></tr><tr><td>Continuous Assessment</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Problem sets</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Written report</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Midterm examination</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Final examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Problem sets – Student will work on practice problems to avoid building up gaps in understanding as they progress through the course. Written report –Students will apply tools and skills acquired in the course to analyze a topic related to course content. Midterm exam – Students will be assessed on their understanding of the first half of knowledge points.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						(1)	(2)	(3)	(4)	(5)		Continuous Assessment								Problem sets	10%	✓	✓	✓	✓	✓		Written report	20%	✓	✓	✓	✓	✓		Midterm examination	20%	✓	✓	✓	✓	✓		Final examination	50%	✓	✓	✓	✓	✓										Total	100%						
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Final examination	50%	✓	✓	✓	✓	✓																																																																	
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	Final exam – students will be assessed on their overall understanding of microeconomic theory and applications.	
Student Study Effort Expected	Class contact:	
	▪ Lectures / Tutorials	39 Hrs.
	Other student study effort:	
	▪ Consultation of textbook and references	39 Hrs.
	▪ Problem solving	39 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	The required textbook is: - Nicholson, W., & Snyder, C. M. (2025). <i>Microeconomic Theory: Basic Principles and Extensions (13th Edition)</i>. Cengage Learning. Students may also wish to consult the following textbooks for discussions at a similar level: - Wetzstein, Michael E. (2013). <i>Microeconomic Theory: Concepts and Connections (2nd Edition)</i>. Routledge. - Varian, Hal (2014). <i>Intermediate Microeconomics (9th Edition)</i>. W. W. Norton. Students interested in more advanced treatments in microeconomic theory may consult: - Kreps, David M. (1990). <i>A Course in Microeconomic Theory</i>. Princeton University Press. - Varian, Hal (1992). <i>Microeconomic Analysis (3rd Edition)</i>. W. W. Norton. Students interested in studying game theory in greater detail may consult: - Osborne, Martin J. (2003). <i>An Introduction to Game Theory</i>. Oxford University Press.	